

ASTROPHOTOGRAPHY

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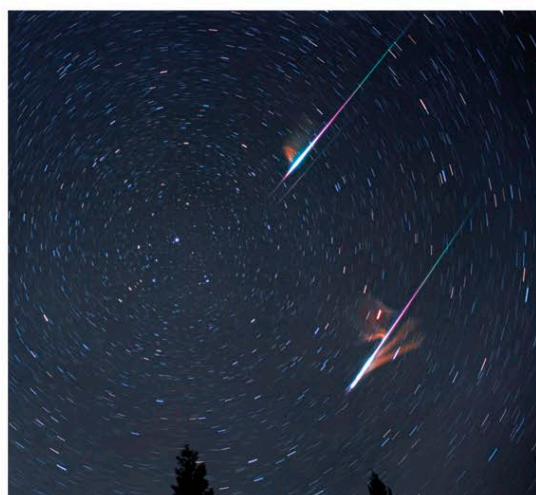
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WITH MODERN TECHNOLOGY, amateur astronomers can now take images that would previously have been possible only from professional observatories. Even compact digital cameras can photograph bright objects, such as the Moon, through a telescope and can capture sky views, such as twilight scenes and constellations.



BASIC ASTROPHOTOGRAPHY

Almost any camera can be used to take pictures of the night sky, although without a telescope it is limited to recording little more than naked-eye views of the stars, the Moon, bright planets, meteor trails, constellations, and aurorae. The main requirement for basic astrophotography is that the camera can keep the shutter open for long periods—at least several seconds. With long exposures, it is essential to keep the camera steady by mounting it on a tripod. Using a cable release, remote release, or timer to trigger the shutter will also help to avoid shake and blurring of the image.



METEORS

Individual meteors cannot be predicted and so the only way to photograph them is to use long exposures in the hope that one will appear by chance. The field of view of an ordinary camera is ideal, and the exposure time should be as long as possible without the image being saturated by background light. Bright meteors will record as streaks against the background of star trails.

FIXED-CAMERA SHOTS

General sky photography requires exposure times of many seconds with the camera at its most sensitive setting and focused on infinity. Mount the camera on a tripod to hold it steady during the exposure.

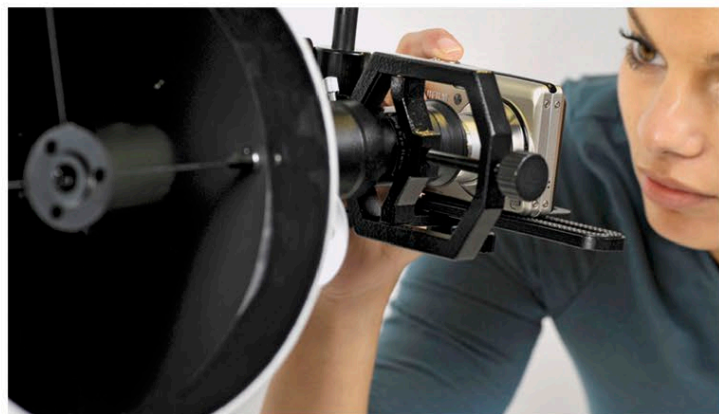


STAR TRAILS

During a fixed-camera exposure of more than a few seconds, the stars will trail across the image as they appear to move due to Earth's rotation—in this case, around the celestial pole. In light-polluted areas, take numerous shorter exposures and stack them using image-processing software to avoid an overexposed sky background.

DIGISCOPING AND PIGGYBACKING

Compact cameras or cell phones can be used to take images directly through a telescope, a technique known as digiscoping. At its simplest, the camera can be mounted on a tripod and pointed down the telescope eyepiece. Alternatively, an adapter can be used to fix the camera to the eyepiece. Attaching the camera on top of a motor-driven equatorially mounted telescope—known as piggybacking—allows long-exposure views of the sky and even deep-sky objects without producing trails on the image. The image recorded is the one captured by the camera, not that seen through the telescope.



DIGISCOPING IMAGE OF THE MOON

Excellent images of the Moon can be obtained by digiscoping with even simple cameras. As the Moon is so bright, the exposure time for a Moon picture is similar to that for an ordinary daytime shot.



DIGISCOPING SET-UP

An adapter enables the camera to be aligned with the telescope eyepiece. Set the camera to manual exposure and use the self-timer to avoid shaking the camera. Experiment with different exposure times for the best results.

piggyback-mounted camera with telephoto lens



PIGGYBACK SET-UP

Many motor-driven telescopes have a threaded bolt for piggybacking a camera. If a telephoto lens is mounted on the camera and a long exposure is used, clear images of even deep-sky objects can be obtained.